## Example 6 (Similar to 3.60c)

Use the tables of transforms and properties to find the DTFT of the following signal

$$x[n] = \left(\frac{3}{7}\right)^n (n+1)\ \sin\left(\frac{\pi}{4}n\right) u[n]$$


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# Example 6

$$x[n] = \left(\frac{3}{7}\right)^n (n+1)\ \sin\left(\frac{\pi}{4}n\right) u[n]$$

Which properties and transform pairs can be used? Talk with your neighbor.

$C.3\ pg\ 774$

$$x[n] = (n+1)\alpha^n u[n], \xleftrightarrow{FT} X(e^{j\Omega}) = \frac{1}{(1-\alpha e^{-j\Omega})^2}$$

C.6, pg. 776

$$\sin(\Omega_1 n) \overset{\mathcal{F}}{\leftrightarrow} \frac{\pi}{j} \sum_{k=-\infty}^{\infty} \delta(\Omega - \Omega_1 - k2\pi) - \delta(\Omega + \Omega_1 - k2\pi)$$

$C.7\ cont. Multiplication\ pg\ 778$

$$x[n]y[n] \xleftrightarrow{FT} \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\Gamma})Y(e^{j(\Omega-\Gamma)})d\Gamma$$

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# Example 6

$$x[n] = \left(\frac{3}{7}\right)^n (n+1)\sin\left(\frac{\pi}{4}n\right)u[n] = \mathrm{a[n]b[n]}$$

$$a[n] = (n+1)\left(\frac{3}{7}\right)^n u[n] \xleftrightarrow{FT} A\left(e^{j\Omega}\right) = \frac{1}{\left(1-\left(\frac{3}{7}\right)e^{-j\Omega}\right)^2}$$

$$b[n] = \sin(\Omega_1 n) \overset{\mathcal{F}}{\leftrightarrow} B\left(e^{j\Omega}\right) = \frac{\pi}{j}\sum_{k=-\infty}^{\infty} \delta(\Omega - \Omega_1 - k2\pi) - \delta(\Omega + \Omega_1 - k2\pi)$$

$$x[n] = a[n]b[n] \overset{\mathcal{F}}{\leftrightarrow} X\left(e^{j\Omega}\right) = \frac{1}{2\pi}\int_{-\pi}^{\pi} A\left(e^{j\Gamma}\right)B\left(e^{j(\Omega-\Gamma)}\right)d\Gamma$$

$$B\left(e^{j(\Omega-\Gamma)}\right) = \frac{\pi}{j}\sum_{k=-\infty}^{\infty} \delta\left(\Omega - \Gamma - \frac{\pi}{4} - k2\pi\right) - \delta\left(\Omega - \Gamma + \frac{\pi}{4} - k2\pi\right)$$


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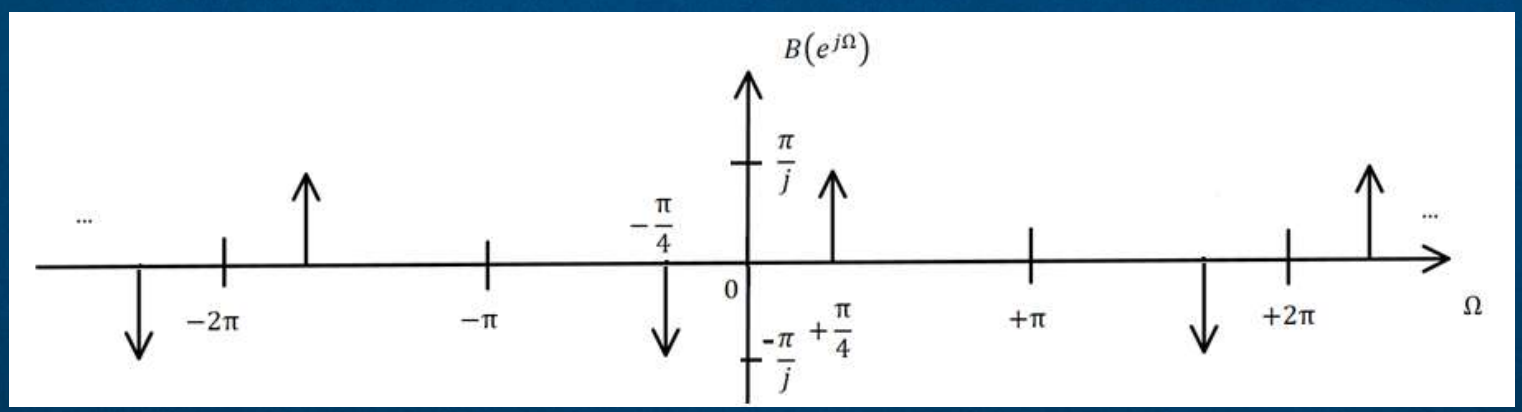
# Example 6

$$b[n] = \sin(\Omega_1 n) \overset{\mathcal{F}}{\leftrightarrow} B\left(e^{j\Omega}\right) = \frac{\pi}{j} \sum_{k=-\infty}^{\infty} \delta(\Omega - \Omega_1 - k2\pi) - \delta(\Omega + \Omega_1 - k2\pi)$$

$$b[n] = \sin\left(\frac{\pi}{4} n\right) \overset{\mathcal{F}}{\leftrightarrow} B\left(e^{j\Omega}\right) = \frac{\pi}{j} \sum_{k=-\infty}^{\infty} \delta\left(\Omega - \frac{\pi}{4} - k2\pi\right) - \delta\left(\Omega + \frac{\pi}{4} - k2\pi\right)$$





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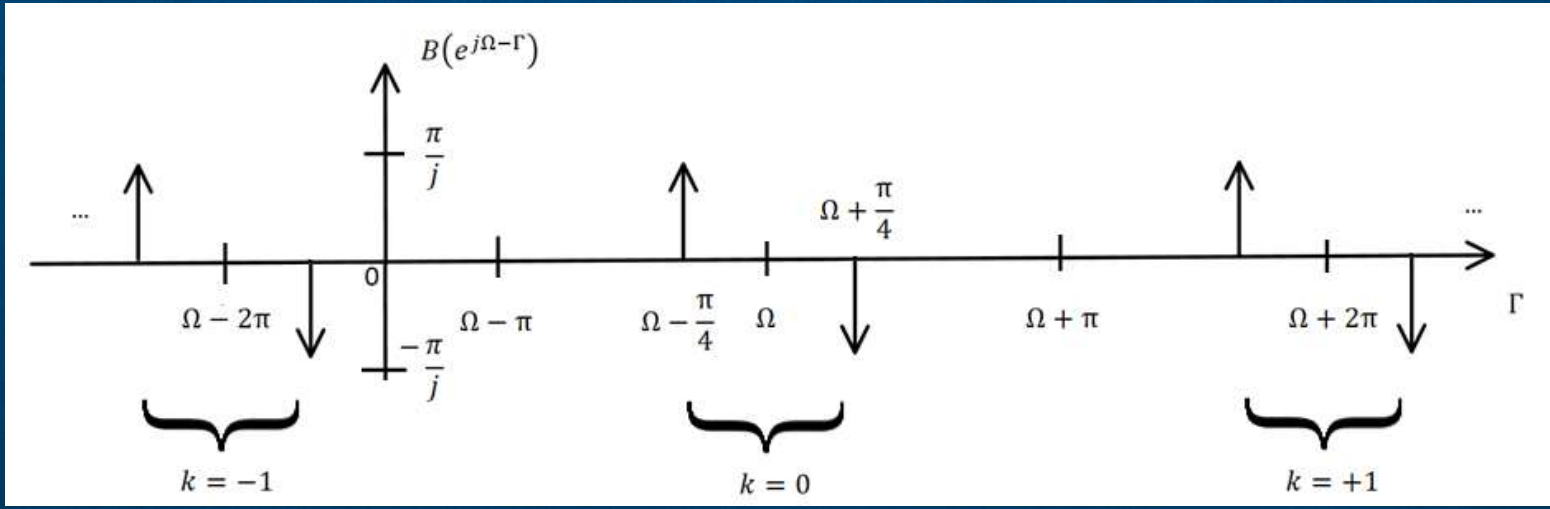
# Example 6

$$B\left(e^{j(\Omega-\Gamma)}\right) = \frac{\pi}{j}\sum_{k=-\infty}^{\infty} \delta\left(\Omega - \frac{\pi}{4} - k2\pi - \Gamma\right) - \delta\left(\Omega + \frac{\pi}{4} - k2\pi - \Gamma\right)$$





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# Example 6

$$X\left(e^{j\Omega}\right) = \frac{1}{2\pi}\int_{-\pi}^{\pi} A\left(e^{j\Gamma}\right)B\left(e^{j(\Omega-\Gamma)}\right)d\Gamma$$

$$X\left(e^{j\Omega}\right) = \frac{1}{2\pi}\int_{\Omega-\pi}^{\Omega+\pi}\left[\left\{\frac{1}{\left(1-\left(\frac{3}{7}\right)e^{-j\Gamma}\right)^2}\right\}\frac{\pi}{j}\sum_{k=-\infty}^{\infty}\delta\left(\Omega-\frac{\pi}{4}-k2\pi-\Gamma\right)-\delta\left(\Omega+\frac{\pi}{4}-k2\pi-\Gamma\right)\right]d\Gamma$$

$$X\left(e^{j\Omega}\right) = \frac{1}{2\pi}\int_{\Omega-\pi}^{\Omega+\pi}\left[\left\{\frac{1}{\left(1-\left(\frac{3}{7}\right)e^{-j\Gamma}\right)^2}\right\}\left\{\frac{\pi}{j}\delta\left(\Omega-\frac{\pi}{4}-\Gamma\right)-\frac{\pi}{j}\delta\left(\Omega+\frac{\pi}{4}-\Gamma\right)\right\}\right]d\Gamma$$

Since

$$f(t_0) = \int_{-\infty}^{\infty} f(t)\delta(t-t_0)dt\,, \qquad \text{(sifting property of an impulse)}$$

$$\boxed{X\left(e^{j\Omega}\right) = \frac{1}{2j\left(1-\left(\frac{3}{7}\right)e^{-j\left(\Omega-\frac{\pi}{4}\right)}\right)^2} - \frac{1}{2j\left(1-\left(\frac{3}{7}\right)e^{-j\left(\Omega+\frac{\pi}{4}\right)}\right)^2}}$$

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## Practice on the boards

3.59(a)
Use the tables of transforms and properties to find the inverse FTs of the following signal:

$$X(j\omega) = \frac{j\omega}{(7 + j\omega)^2}$$


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3.59(a)
Use the tables of transforms and properties to find the inverse FTs of the following signal:

$$X(j\omega) = \frac{j\omega}{(7 + j\omega)^2}$$

C.4 (pg. 775)

$$x(t) = te^{-at}u(t), Re\{a\} > 0 \xleftrightarrow{FT} X(j\omega) = \frac{1}{(a + j\omega)^2}$$

C.7 (pg. 777)

$$x(t) = \frac{d}{dt}x(t) \xleftrightarrow{FT} X(j\omega) = j\omega X(j\omega)$$

$$x(t) = \frac{d}{dt}\{te^{-7t}u(t)\}$$

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## Example 6 (Similar to 3.60c)

Use the tables of transforms and properties to find the DTFT of the following signal

$$x[n] = \left(\frac{3}{7}\right)^n (n+1)\ \sin\left(\frac{\pi}{4}n\right) u[n-3]$$

Which properties and transform pairs can be used?

$$C.3\ pg\ 774$$

$$x[n] = (n+1)\alpha^n u[n], \xleftrightarrow{FT} X(e^{j\Omega}) = \frac{1}{(1-\alpha e^{-j\Omega})^2}$$

Our function doesn't match that exactly because of the $u[n-3]$, but we can make it match…How?

$$C.7\ pg\ 778$$

$$x[n-n_0], \xleftrightarrow{FT} e^{-j\Omega n_0} X(e^{j\Omega})$$


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# Example 6

$$x[n] = \left(\frac{3}{7}\right)^n (n+1)\sin\left(\frac{\pi}{4}n\right)u[n-3] = w[n]\sin\left(\frac{\pi}{4}n\right)$$

$$w[n] = \left(\frac{3}{7}\right)^n (n+1)u[n-3]$$

$$= \left(\frac{3}{7}\right)^3 \left(\frac{3}{7}\right)^{n-3} (n+1)u[n-3]$$

$$= \overbrace{\left(\frac{3}{7}\right)^3 \left(\frac{3}{7}\right)^{n-3} (n-3+1)u[n-3]}^{a[n-3]} + \overbrace{3\left(\frac{3}{7}\right)^3 \left(\frac{3}{7}\right)^{n-3} u[n-3]}^{b[n-3]}$$

$$= a[n-3] + b[n-3]$$

where

$$a[n] = \left(\frac{3}{7}\right)^3 \left(\frac{3}{7}\right)^n (n+1)u[n]$$

$$b[n] = 3\left(\frac{3}{7}\right)^3 \left(\frac{3}{7}\right)^n u[n].$$

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# Example 6

$$x[n] = w[n]\sin\left(\frac{\pi}{4}n\right) = (a[n-3] + b[n-3])\sin\left(\frac{\pi}{4}n\right)$$

$$a[n] = \left(\frac{3}{7}\right)^3\left(\frac{3}{7}\right)^n(n+1)u[n]; \quad b[n] = 3\left(\frac{3}{7}\right)^3\left(\frac{3}{7}\right)^n u[n].$$

Which properties and transform pairs can be used?

C.6, pg. 776

$$\sin(\Omega_1 n) \overset{\mathcal{F}}{\leftrightarrow} \frac{\pi}{j}\sum_{k=-\infty}^{\infty}\delta(\Omega - \Omega_1 - k2\pi) - \delta(\Omega + \Omega_1 - k2\pi)$$

$C.3\ pg\ 774$

$$x[n] = (n+1)\alpha^n u[n], \overset{FT}{\longleftrightarrow} X(e^{j\Omega}) = \frac{1}{(1-\alpha e^{-j\Omega})^2}$$

$C.3\ pg\ 774$

$$x[n] = \alpha^n u[n], |\alpha| < 1 \overset{FT}{\longleftrightarrow} X(e^{j\Omega}) = \frac{1}{1-\alpha e^{-j\Omega}}$$

$C.7\ pg\ 778$

$$x[n-n_0], \quad \overset{FT}{\longleftrightarrow} \quad e^{-j\Omega n_0}X(e^{j\Omega})$$

$C.7$ cont. Multiplication pg 778

$$x[n]y[n] \overset{FT}{\longleftrightarrow} \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\Gamma})Y(e^{j(\Omega-\Gamma)})d\Gamma$$

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# Example 6

$$C.3\ pg\ 774$$

$$x[n] = (n+1)\alpha^n u[n] \xleftrightarrow{FT} X(e^{j\Omega}) = \frac{1}{(1-\alpha e^{-j\Omega})^2}$$

$$a[n] = \left(\frac{3}{7}\right)^3 \left(\frac{3}{7}\right)^n (n+1)u[n] \xleftrightarrow{FT} A(e^{j\Omega}) = \left(\frac{3}{7}\right)^3 \frac{1}{\left(1-\frac{3}{7}e^{-j\Omega}\right)^2}$$

$$C.3\ pg\ 774$$

$$x[n] = \alpha^n u[n], |\alpha| < 1 \xleftrightarrow{FT} X(e^{j\Omega}) = \frac{1}{1-\alpha e^{-j\Omega}}$$

$$b[n] = 3\left(\frac{3}{7}\right)^3 \left(\frac{3}{7}\right)^n u[n] \xleftrightarrow{FT} B(e^{j\Omega}) = 3\left(\frac{3}{7}\right)^3 \frac{1}{1-\frac{3}{7}e^{-j\Omega}}$$

$$C.7\ pg\ 778$$

$$x[n-n_0] \xleftrightarrow{FT} e^{-j\Omega n_0} X(e^{j\Omega})$$

$$w[n] = a[n-3] + b[n-3] \xleftrightarrow{FT} e^{-j\Omega 3}\left(\frac{3}{7}\right)^3 \frac{1}{\left(1-\frac{3}{7}e^{-j\Omega}\right)^2} + 3e^{-j\Omega 3}\left(\frac{3}{7}\right)^3 \frac{1}{1-\frac{3}{7}e^{-j\Omega}}$$


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## Example 6

$$W(e^{j\Omega}) = e^{-j\Omega 3}\left(\frac{3}{7}\right)^3 \frac{1}{\left(1-\frac{3}{7}e^{-j\Omega}\right)^2} + 3e^{-j\Omega 3}\left(\frac{3}{7}\right)^3 \frac{1}{1-\frac{3}{7}e^{-j\Omega}}$$

$$W(e^{j\Omega}) = \left(\frac{3}{7}\right)^3 e^{-j3\Omega}\left(\frac{1}{\left(1-\frac{3}{7}e^{-j\Omega}\right)^2} + \frac{3}{1-\frac{3}{7}e^{-j\Omega}}\right)$$

C.6, pg. 776

$$\sin(\Omega_1 n) \overset{\mathcal{F}}{\leftrightarrow} \frac{\pi}{j}\sum_{k=-\infty}^{\infty} \delta(\Omega-\Omega_1-k2\pi) - \delta(\Omega+\Omega_1-k2\pi)$$

$$y[n] = \sin\left(\frac{\pi}{4}n\right) \overset{\mathcal{FT}}{\leftrightarrow} Y(e^{j\Omega}) = \frac{\pi}{j}\sum_{k=-\infty}^{\infty} \delta\left(\Omega-\frac{\pi}{4}-k2\pi\right) - \delta\left(\Omega+\frac{\pi}{4}-k2\pi\right)$$


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# Example 6

$$x[n] = w[n]\overbrace{\sin\left(\frac{\pi}{4}n\right)}^{y[n]}$$

$C.7$ cont. Multiplication pg 778

$$x[n]y[n] \xleftrightarrow{FT} \frac{1}{2\pi}\int_{-\pi}^{\pi} X\left(e^{j\Gamma}\right)Y\left(e^{j(\Omega-\Gamma)}\right)d\Gamma$$

$$W\left(e^{j\Gamma}\right) = \left(\frac{3}{7}\right)^3 e^{-j3\Gamma}\left(\frac{1}{\left(1-\frac{3}{7}e^{-j\Gamma}\right)^2} + \frac{3}{1-\frac{3}{7}e^{-j\Gamma}}\right)$$

$$Y\left(e^{j\Omega-\Gamma}\right) = \frac{\pi}{j}\sum_{k=-\infty}^{\infty}\delta\left(\Omega-\Gamma-\frac{\pi}{4}-k2\pi\right) - \delta\left(\Omega-\Gamma+\frac{\pi}{4}-k2\pi\right)$$

$$X\left(e^{j\Omega}\right) = \frac{1}{2\pi}\int_{-\pi}^{\pi} W\left(e^{j\Gamma}\right)\frac{\pi}{j}\sum_{k=-\infty}^{\infty}\left(\delta\left(\Omega-\frac{\pi}{4}-k2\pi-\Gamma\right) - \delta\left(\Omega+\frac{\pi}{4}-k2\pi-\Gamma\right)\right)d\Gamma$$


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# Example 6

$$X(e^{j\Omega}) = \frac{1}{2\pi}\int_{-\pi}^{\pi} W(e^{j\Gamma})\frac{\pi}{j}\sum_{k=-\infty}^{\infty}\left(\delta\left(\Omega - \frac{\pi}{4} - k2\pi - \Gamma\right) - \delta\left(\Omega + \frac{\pi}{4} - k2\pi - \Gamma\right)\right)d\Gamma$$

Using $\int_{-\infty}^{\infty} f(t)\delta(t-t_0)dt = f(t_0)$

$$X(e^{j\Omega}) = \frac{1}{2j}\left(W\left(e^{j\left(\Omega-\frac{\pi}{4}\right)}\right) - W\left(e^{j\left(\Omega+\frac{\pi}{4}\right)}\right)\right)$$

$$W(e^{j\Gamma}) = \left(\frac{3}{7}\right)^3 e^{-j3\Gamma}\left(\frac{1}{\left(1-\frac{3}{7}e^{-j\Gamma}\right)^2} + \frac{3}{1-\frac{3}{7}e^{-j\Gamma}}\right)$$

$$\boxed{X(e^{j\Omega}) = \frac{1}{2j}\left(\frac{3}{7}\right)^3\left(e^{-j3\left(\Omega-\frac{\pi}{4}\right)}\left(\frac{1}{\left(1-\frac{3}{7}e^{-j\left(\Omega-\frac{\pi}{4}\right)}\right)^2} + \frac{3}{1-\frac{3}{7}e^{-j\left(\Omega-\frac{\pi}{4}\right)}}\right) - e^{-j3\left(\Omega+\frac{\pi}{4}\right)}\left(\frac{1}{\left(1-\frac{3}{7}e^{-j\left(\Omega+\frac{\pi}{4}\right)}\right)^2} + \frac{3}{1-\frac{3}{7}e^{-j\left(\Omega+\frac{\pi}{4}\right)}}\right)\right)}$$


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# Multiplication Property

Similarly, if $x[n]$ and $z[n]$ are discrete-time nonperiodic signals, then the DTFT of the product $y[n] = x[n]z[n]$ is given by the convolution of the their DTFT's and multiplication by $1/(2\pi)$; that is,

$$\boxed{y[n] = x[n]z[n] \xleftrightarrow{DTFT} Y(e^{j\Omega}) = \frac{1}{2\pi} X(e^{j\Omega}) \circledast Z(e^{j\Omega}),} \tag{3.57}$$

where, as before, the symbol $\circledast$ denotes periodic convolution. Here, $X(e^{j\Omega})$ and $Z(e^{j\Omega})$ are $2\pi$-periodic, so we evaluate the convolution over a $2\pi$ interval:

$$X(e^{j\Omega}) \circledast Z(e^{j\Omega}) = \int_{-\pi}^{\pi} X(e^{j\theta}) Z(e^{j(\Omega-\theta)})\, d\theta.$$


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# The sine cardinal (sinc) function: $\mathrm{sinc}(x) \equiv \frac{\sin(x)}{x}$

$$\mathrm{sinc}(x) \equiv \frac{\sin(x)}{x}$$

What is the value of the sinc function at the origin?

$$\mathrm{sinc}(0) = \lim_{x\to 0} \frac{\sin(x)}{x} = \lim_{x\to 0} \frac{\frac{d}{dx}\sin(x)}{\frac{d}{dx}x} = \lim_{x\to 0} \frac{\cos(x)}{1}$$

$$\boxed{\mathrm{sinc}(0) = 1}$$

At what values of x does the sinc function evaluate to 0?

$$\mathrm{sinc}(x) = \frac{\sin(x)}{x} = 0$$

when

$$\boxed{x = m\pi, \qquad m = \pm 1, \pm 2, \ldots}$$

What is the area under the sinc function from $-\infty$ to $\infty$?

$$\int_{-\infty}^{\infty} \mathrm{sinc}(x)\, dx = \pi$$

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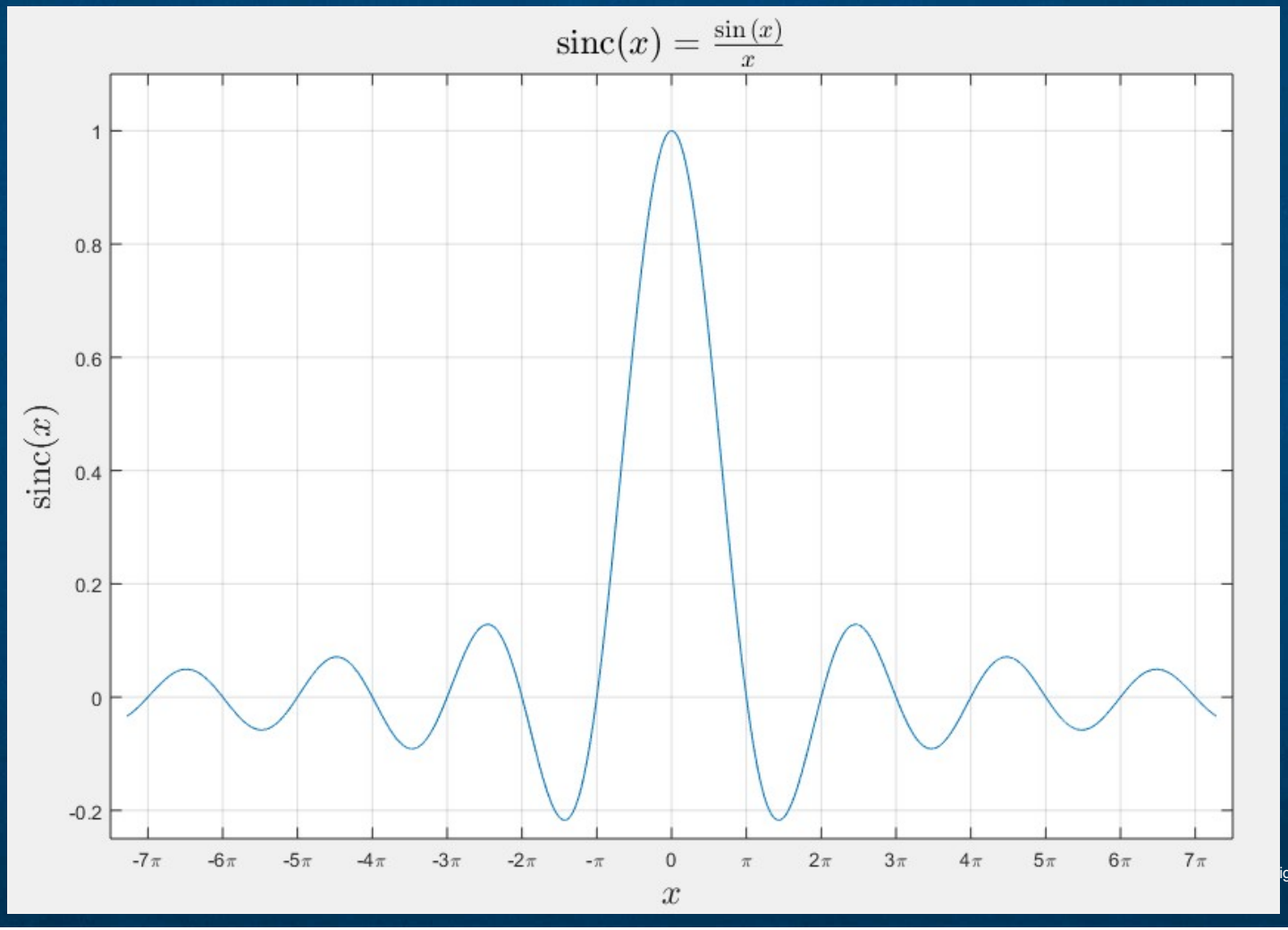
# The sinc function: $\mathrm{sinc}(x) \equiv \frac{\sin(x)}{x}$



# The sine cardinal (sinc) function: $\frac{A}{B}\operatorname{sinc}(Ax) = \frac{\sin(Ax)}{Bx}$

$$\frac{A}{B}\operatorname{sinc}(Ax) = \frac{A}{B}\left(\frac{\sin(Ax)}{(Ax)}\right) = \frac{\sin(Ax)}{Bx}, \qquad x, A, B \in \mathbb{R}$$

What is the value of the sinc function at the origin?

$$\frac{\mathrm{A}}{B}\operatorname{sinc}(0) = \lim_{x\to 0}\frac{\sin(Ax)}{Bx} = \lim_{x\to 0}\frac{\frac{d}{dx}\sin(Ax)}{\frac{d}{dx}Bx} = \lim_{x\to 0}\frac{A\cos(Ax)}{B}$$

$$\boxed{\frac{A}{B}\operatorname{sinc}(0) = \frac{A}{B}}$$

At what values of x does the sinc function evaluate to 0?

$$\frac{A}{B}\operatorname{sinc}(Ax) = \frac{\sin(Ax)}{Bx} = 0$$

when

$$Ax = m\pi, \qquad m = \pm 1, \pm 2, \ldots$$

Thus, $\frac{\mathrm{A}}{\mathrm{B}}\operatorname{sinc}(Ax) = \frac{\sin(Ax)}{Bx} = 0$ when

$$\boxed{x = \frac{m\pi}{A}, \qquad m = \pm 1, \pm 2, \ldots}$$

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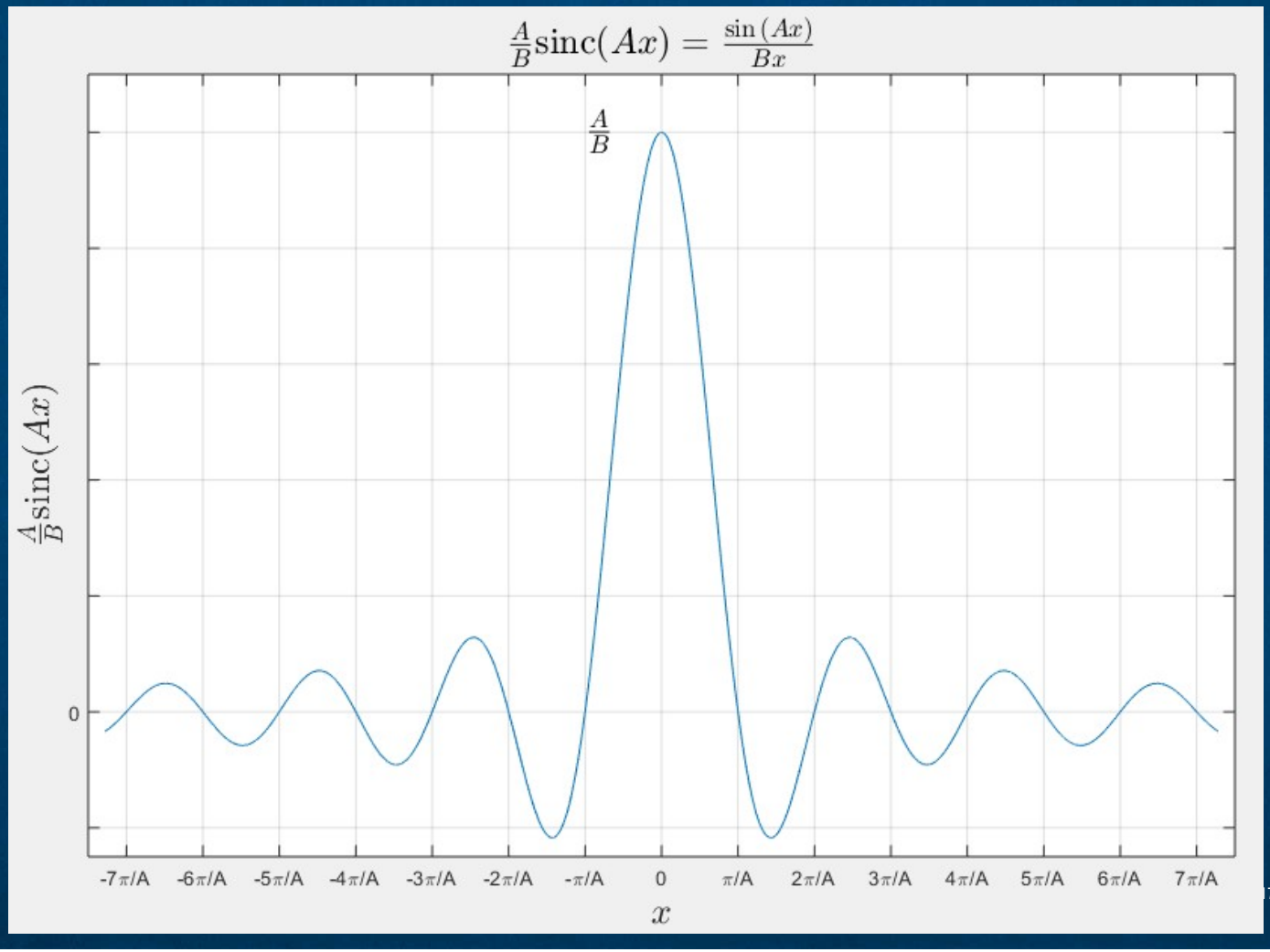
# The sinc function: $\frac{\mathrm{A}}{\mathrm{B}}\mathrm{sinc}(Ax) \equiv \frac{\sin(Ax)}{Bx}$



## The sine cardinal (sinc) function: $\frac{A}{B}\text{sinc}(Ax) = \frac{\sin(Ax)}{Bx}$

$$\frac{A}{B}\text{sinc}(Ax) = \frac{\sin(Ax)}{Bx}$$

where

$$x, A, B \in \mathbb{R}$$

What is the derivative of the sinc function?

$$\frac{d}{dx}\left(\frac{A}{B}\text{sinc}(Ax)\right) = \frac{A}{B}\left(\frac{A\cos(Ax)}{x} - \frac{\sin(Ax)}{x^2}\right)$$

Where are the local extrema (min/max values) for this sinc function?

$$\frac{A\cos(Ax)}{x} - \frac{\sin(Ax)}{x^2} = 0$$

$$\boxed{\cos(Ax) = \frac{\sin(Ax)}{Ax} = \text{sinc}(Ax)}$$

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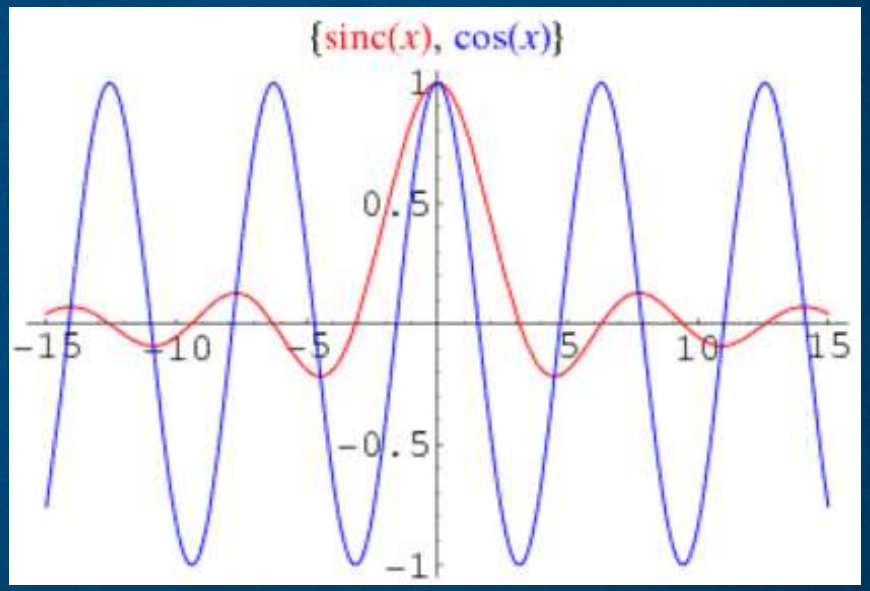
The sine cardinal (sinc) function: $\frac{A}{B}\mathrm{sinc}(Ax) = \frac{\sin(Ax)}{Bx}$

Where are the local extrema (min/max values) for this sinc function?

$$\boxed{\frac{A}{B}\cos(Ax) = \frac{A}{B}\left(\frac{\sin(Ax)}{Ax}\right) = \frac{A}{B}\mathrm{sinc}(Ax)}$$

An interesting property of $\mathrm{sinc}(Ax)$ is that the set of local extrema of $\mathrm{sinc}(Ax)$ corresponds to its intersections with the cosine function $\cos(Ax)$





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# Duality of rectangle pulses and sinc functions

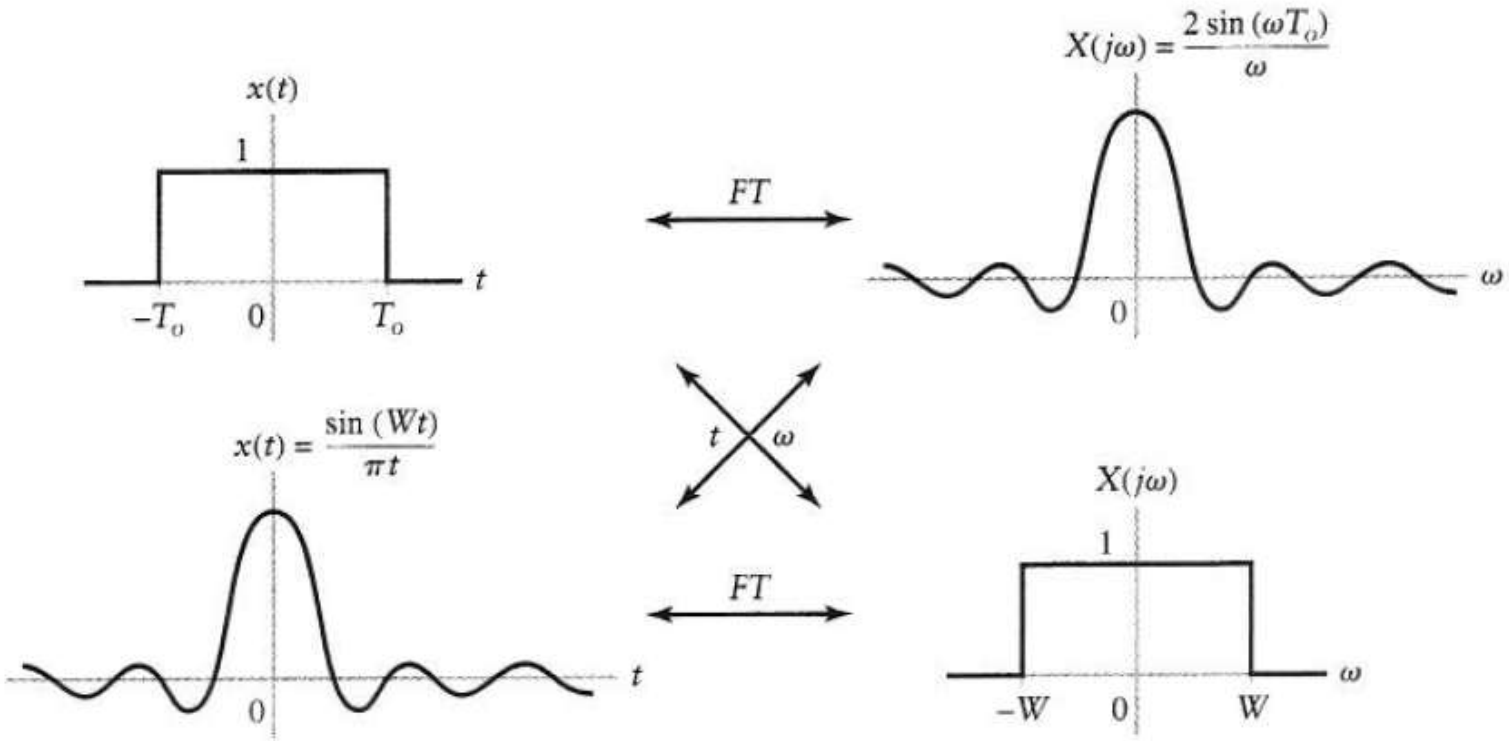


**FIGURE 3.73** Duality of rectangular pulses and sinc functions.


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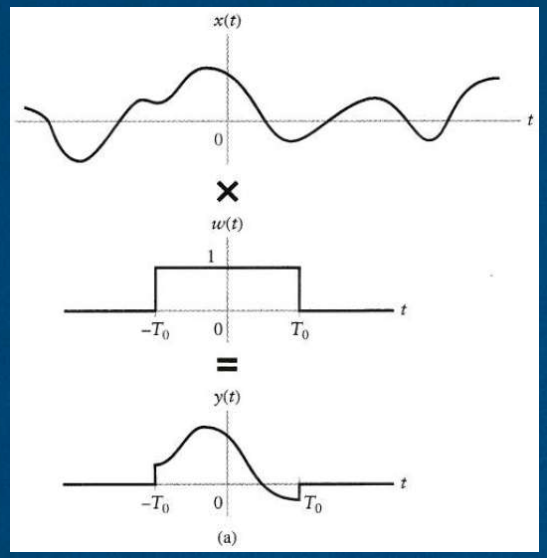
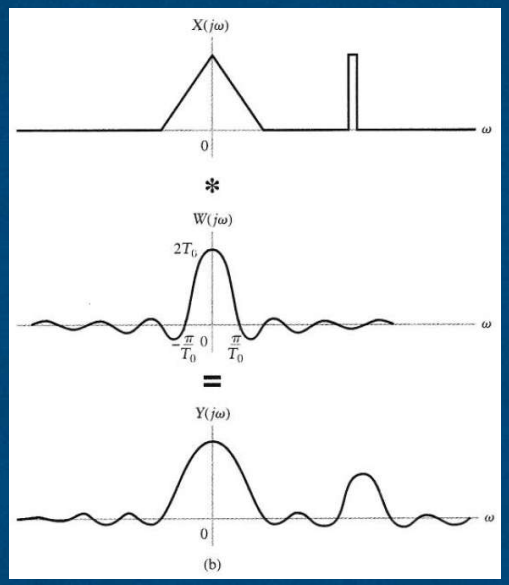
# Windowing




$$W(j\omega) = \frac{2}{\omega}\sin(\omega T_o)$$

$$y(t) = x(t)w(t) \quad \xleftrightarrow{FT} \quad Y(j\omega) = \frac{1}{2\pi}X(j\omega) * W(j\omega)$$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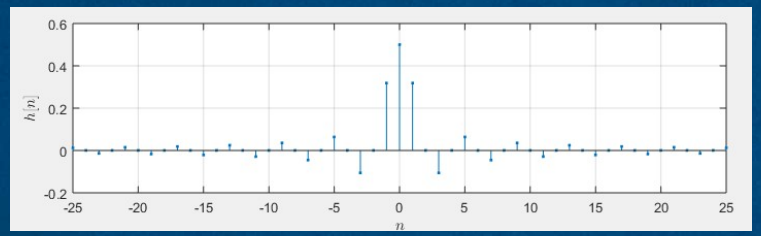
# Example 3.46 (1 of 4)

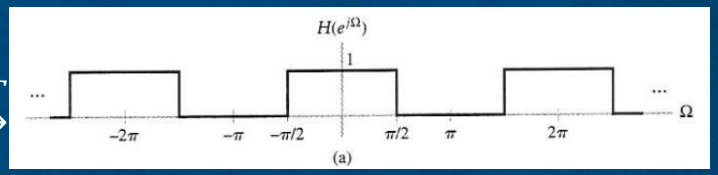
**EXAMPLE 3.46 TRUNCATING THE IMPULSE RESPONSE** The frequency response $H(e^{j\Omega})$ of an ideal discrete-time system is depicted in Fig. 3.66(a). Describe the frequency response of a system whose impulse response is the ideal system impulse response truncated to the interval $-M \le n \le M$.

Ideal system Impulse Response

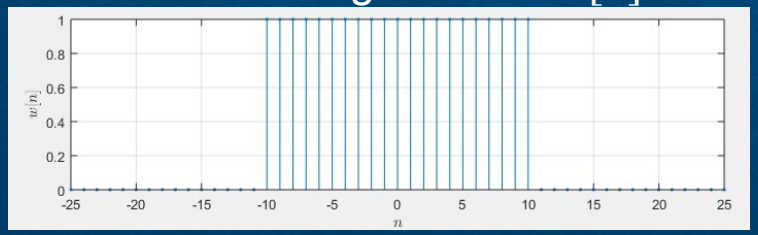



$\xleftrightarrow{DTFT}$

Fig. 3.66(a) Frequency Response $H(e^{j\Omega})$

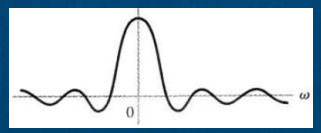



Windowing function $w[n]$

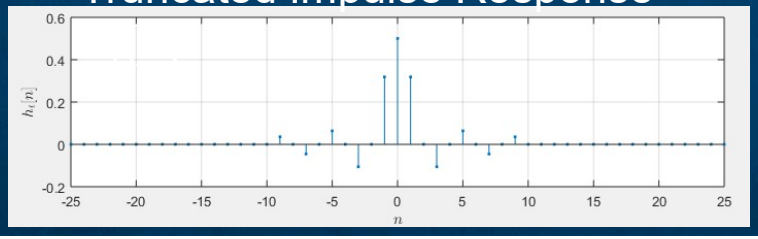



$\xleftrightarrow{DTFT}$

$W(e^{j\Omega})$

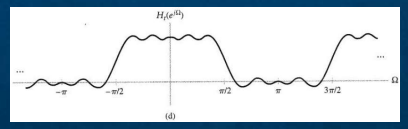



Truncated Impulse Response




$\xleftrightarrow{DTFT}$

Describe this:
Frequency Response $H_t(e^{j\Omega})$



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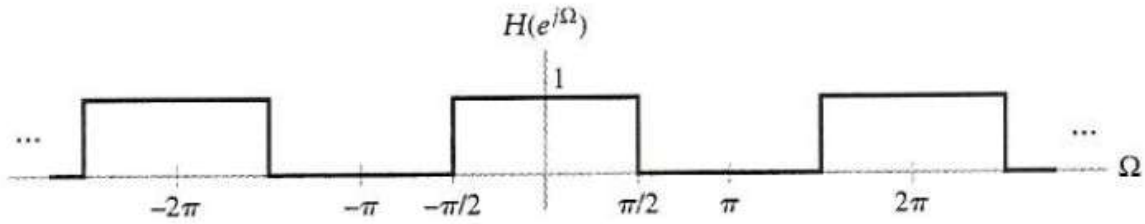
# Example 3.46 (2 of 4)

**EXAMPLE 3.46 TRUNCATING THE IMPULSE RESPONSE** The frequency response $H(e^{j\Omega})$ of an ideal discrete-time system is depicted in Fig. 3.66(a). Describe the frequency response of a system whose impulse response is the ideal system impulse response truncated to the interval $-M \le n \le M$.



(a)

Using

$$X(e^{j\Omega}) = \begin{cases} 1, & |\Omega| \le W \\ 0, & W < |\Omega| \le \pi \end{cases} \quad \xleftrightarrow{DTFT} \quad x[n] = \begin{cases} \dfrac{1}{\pi n}\sin(Wn), & n \ne 0 \\ \dfrac{W}{\pi}, & n = 0 \end{cases}$$

then

$$H(e^{j\Omega}) = \begin{cases} 1, & |\Omega| \le \dfrac{\pi}{2} \\ 0, & \dfrac{\pi}{2} < |\Omega| \le \pi \end{cases} \quad \xleftrightarrow{DTFT} \quad h[n] = \begin{cases} \dfrac{1}{\pi n}\sin\left(\dfrac{\pi}{2}n\right), & n \ne 0 \\ \dfrac{1}{2}, & n = 0 \end{cases}.$$

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This response is infinite in extent. Let $h_t[n]$ be the truncated impulse response:

$$h_t[n] = \begin{cases} h[n], & |n| \le M \\ 0, & \text{otherwise} \end{cases}.$$

We may express $h_t[n]$ as the product of $h[n]$ and a window function $w[n]$, where

$$w[n] = \begin{cases} 1, & |n| \le M \\ 0, & \text{otherwise} \end{cases}.$$

Let $h_t[n] \xleftrightarrow{DTFT} H_t(e^{j\Omega})$, and use the multiplication property given by Eq. (3.57) to obtain

$$H_t(e^{j\Omega}) = \frac{1}{2\pi}\int_{-\pi}^{\pi} H(e^{j\theta})W(e^{j(\Omega-\theta)})\,d\theta.$$

Since

$$H(e^{j\theta}) = \begin{cases} 1, & |\theta| < \pi/2 \\ 0, & \pi/2 < |\theta| < \pi \end{cases},$$

and because, on the basis of Example 3.18, we have

$$W(e^{j(\Omega-\theta)}) = \frac{\sin((\Omega-\theta)(2M+1)/2)}{\sin((\Omega-\theta)/2)},$$

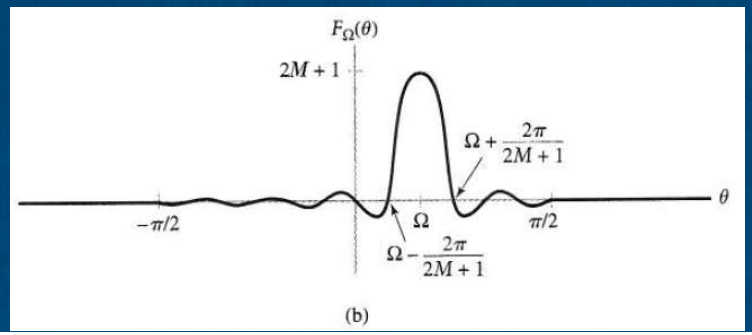
it follows that

$$H_t(e^{j\Omega}) = \frac{1}{2\pi}\int_{-\pi/2}^{\pi/2} F_\Omega(\theta)\,d\theta,$$

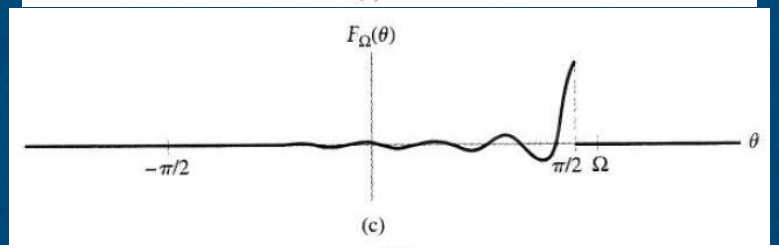
where we have defined

$$F_\Omega(\theta) = H(e^{j\theta})W(e^{j(\Omega-\theta)}) = \begin{cases} W(e^{j(\Omega-\theta)}), & |\theta| < \pi/2 \\ 0, & |\theta| > \pi/2 \end{cases}.$$

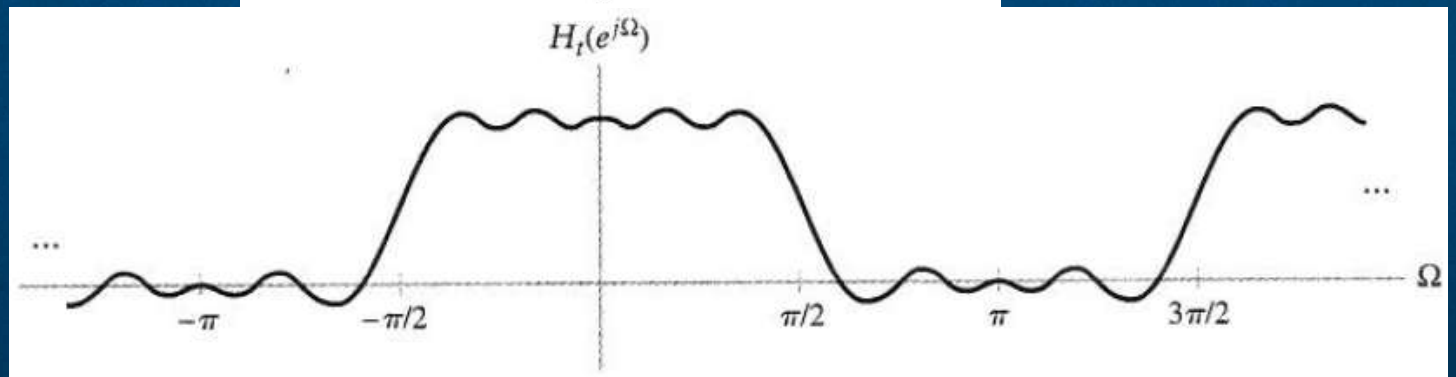
# Example 3.46 (4 of 4)




(b)




(c)




(d)

# Example 7 (MATLAB 3.112)

3.112 Evaluate the frequency response of the truncated filter in Example 3.46. You can do this in MATLAB by writing an m-file to evaluate

$$\mathrm{H}_t(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} \mathrm{h}_t[n]e^{-j\Omega n} = \sum_{n=-\infty}^{\infty} \mathrm{h}[n]w[n]e^{-j\Omega n} = \sum_{n=-M}^{M} \mathrm{h}[n]e^{-j\Omega n},$$

where

$$w[n] = \begin{cases} 1, & |n| \le M \\ 0, & \text{otherwise} \end{cases}.$$

for a large number ( $> 1000$) of values of $\Omega$ in the interval $-\pi < \Omega \le \pi$. Plot the frequency response magnitude in dB ($20 \log_{10}|H_t(e^{j\Omega})|$) for the following values of $M$:

(a) $M = 4$

(b) $M = 10$

(c) $M = 25$

(d) $M = 50$

Discuss the effect of increasing $M$ on the accuracy with which $H_t(e^{j\Omega})$ approximates $H(e^{j\Omega})$.



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# Example 7 MATLAB script for MATLAB 3.112

$$\mathrm{H}_t(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} \mathrm{h}_t[n]e^{-j\Omega n}$$

$$= \sum_{n=-\infty}^{\infty} \mathrm{h}[n]w[n]e^{-j\Omega}$$

$$= \sum_{n=-M}^{M} \mathrm{h}[n]e^{-j\Omega n},$$

$$H(e^{j\Omega}) = \begin{cases} 1, & |\Omega| \le \frac{\pi}{2} \\ 0, & \frac{\pi}{2} < |\Omega| \le \pi \end{cases}$$

$$\xleftrightarrow{DTFT}$$

$$h[n] = \begin{cases} \frac{1}{\pi n}\sin\left(\frac{\pi}{2}n\right), & n \ne 0 \\ \frac{1}{2}, & n = 0 \end{cases}.$$

```
% MATLAB script for MATLAB 3.112
clear
M = [4,10,25,50];  % M = 166 makes a cool logo in dB plot
for k = 1:4
Ht = 0;
Omega = -pi:pi/1000:pi;
    for n = -M(k):M(k)
        if n ~= 0
            hn = 1/pi/n*sin(pi*n/2);
        else
            hn = 1/2;
        end
        Ht = Ht + hn*exp(-1j*Omega*n);
    end
    Htmag = abs(Ht);
    HtdB = 20*log10(abs(Ht));
    figure(1)
    subplot(4,2,2*k-1)
    plot(Omega, HtdB)
    grid on
    xlabel('$\Omega$','interpreter','latex')
    y_str = sprintf('%s%i%s','$\left | H_t(e^{j\Omega}) \right |_{dB}, M=',M(k),'$');
    ylabel(y_str,'Interpreter','latex')

    subplot(4,2,2*k)
    plot(Omega, Htmag)
    grid on
    xlabel('$\Omega$','interpreter','latex')
    y_str = sprintf('%s%i%s','$|H_t(e^{j\Omega})|, M=',M(k),'$');
    ylabel(y_str,'Interpreter','latex')
end
```


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